

Big Data Processing Assignment
Deadline: 18.04.25, 11.55 pm IST
Marks: 10

You are given an undirected unweighted graph as a two column file with node ids (integers). Your task is to write a **spark program** to produce all heavy-hitter triangles in the graph. A heavy-hitter triangle is a three node tuple such that they all have edges with one another and each of the nodes is also heavy hitter node.

Input file format: The graph will in a two column file of node ids, where one line denotes one edge. A sample file is given for your reference in moodle.

Output file format: You should produce the output as a three column file, where each line will represent a triangle.

About your program: You can write the spark code either in scala or in python (pyspark). The program must have a main function. I have also uploaded some example program with main function for your ready reference.

How will we run your program for evaluation: We will evaluate your program from command line with the arguments as follows:

spark-submit <your-code> <input file> <output file name>

Submission guidelines: You need to submit the program as a single python file in moodle. The file name must follow the format: **assignment-roll.py /scala** (where the roll denotes your roll number in capital letters that must match exactly with your IITKGP roll number).

Important notes:

1. No credit will be given if your program does not run and produces wrong output.
2. No credit will be given if it is not a spark program.
3. No submission will be accepted after deadline.
4. It is your responsibility to check that the file has been submitted successfully.
5. Plagiarism is not allowed.